O-I-561b
May 16, 1966
SUPERSEDING
Fed. Spec. O-I-561a
August 3, 1962

FEDERAL SPECIFICATION

INSECTICIDE, DDT-LINDANE INDOOR FOGGING TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type and grade of liquid DDT-lindane insecticide (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 **Specifications and standards.** The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

O-I-514—Insecticide, DDT.
O-T-236—Tetrachloroethylene (Perchloroethylene); Technical Grade.
TT-I-558—Ink, Marking Stencil, Opaque For Nonporous Surfaces (Metals, Glass, etc.).
PPP-P-704—Pails: Shipping, Steel (1 Through 12 Gallon).

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Test Method Std. No. 791—Lubricants Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-2104—Lubricating Oil, Internal Combustion Engine, Heavy-Duty.
MIL-A-3941—Adhesive, Paper Label, Water-Resistant.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 6840

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

Federal Insecticide, Fungicide, and Rodenticide Act.

Regulation for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act, Part 362 of Chapter III.

(Application for copies should be addressed to the U. S. Department of Agriculture, Agricultural Research Service, Pesticides Regulation Division, Washington, D. C., 20250.)

Official Methods of Analysis of Association of Official Agricultural Chemists, Ninth Edition (1960).

(Application for copies should be addressed to the Association of Official Analytical Chemists, P. O. Box 540, Benjamin Franklin Station, Washington 4, D. C.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Dichlorodiphenyltrichloroethane (DDT).* The DDT shall conform to O-I-514.

3.1.2 *Lindane.* Lindane shall be a white crystalline substance, the gamma isomer of 1, 2, 3, 4, 5, 6—hexachlorocyclohexane, of a purity not less than 99 percent. It shall have a melting point of 112°-113°C. when determined as specified in 4.3.3.

3.1.3 *Tetrachloroethylene.* The tetrachloroethylene shall conform to O-T-236.

3.1.4 *Lubricating oil.* Lubricating oil shall conform to SAE grade 50 specified in MIL-L-2104.

3.2 Composition. The finished insecticide shall consist of the following components in the proportions specified in table I when tested as specified in 4.3.3.

TABLE I *Components*

Ingredient	Percent by weight
DDT	10 ± 0.5
Lindane	2 ± 0.1
Tetrachloroethylene	83 ± 0.5
Lubricating oil, SAE 50	5 ± 0.3

3.2.1 *Properties.*

3.2.1.1 *Appearance.* The finished insecticide shall have a color reading of not more than 2.0 on the Union colorimeter when tested as specified in 4.3.3.

3.2.1.2 *Water content.* The water content of the finished insecticide shall not exceed 0.15 percent by weight when determined as specified in 4.3.3.

3.3 Labeling data. The labeling data shall be as follows:

INSECTICIDE DDT-LINDANE
INDOOR FOGGING TYPE
FSN

Active ingredients	*Percent by weight*
DDT	10.0
Lindane	2.0
Inert ingredients	
Tetrachloroethylene	83.0
Lubricating oil (SAE grade 50)	5.0
Total	100

MANUFACTURER'S NAME AND ADDRESS
DATE OF PACK
BATCH NO.
CONTRACT NO.
NET CONTENTS

CAUTION

WARNING:

Use extreme care in handling this insecticide.

Vapor harmful
Harmful if swallowed
Absorbed thru skin

Avoid breathing vapor or spray mist. Avoid contact with clothing, skin and eyes.

Avoid contamination of feed and foodstuffs. Under no circumstances should this formulation be used in food-storage areas unless food is packaged in metal or glass containers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

In case of contact, remove contaminated clothing, wash skin with plenty of soap and water; for eyes, flush with water for 15 minutes and obtain medical attention.

Do not enter areas where this aerosol is being applied or while it remains in the air. If it becomes necessary to enter such an area, wear a gas mask approved by the Department of Agriculture for mixing or applying insecticides in confined or inadequately ventilated spaces. Change clothing and shower after leaving.

Note: This formulation tends to soften the rubber hose of some spray machines. After the solution is drained, the hose tends to regain its normal hardness. Occasional use of the insecticide will probably not cause hose trouble. If it is to be used frequently, hose resistant to tetrachloroethylene should be installed.

DIRECTIONS FOR USE

This insecticide formulation is nonexplosive and is to be used for indoor aerosol treatment of warehouses and other buildings. It is not intended for outdoor aerosol operations.

Apply this insecticide with equipment which produces aerosol sprays. For best results, the aerosol particle size should be 5 microns mass median diameter. Particles of this size will not form a wet deposit on a surface passed across the front of the nozzle. Apply at the rate of 1/2 gallon per 100,000 cubic feet of space.

TREATMENT PROCEDURE

Close all doors except one opening to the outside. Close roof ventilators and see that there are no broken panes in windows or skylights. Post warning signs on closed doors. The following wording is suggested:

WARNING

Under aerosol treatment
Do not open door
Do not enter

Treat buildings in late afternoon and leave closed overnight. Since the aerosol resembles smoke, give prior warning to fire officials that the building will be under treatment. Introduce aerosol through open door. When application to a given area is completed, close the outside door and place warning signs on all doors. The morning after treatment, open all outside doors and leave open for at least 30 minutes to air out space.

3.4 **Workmanship.** The finished insecticide shall be a clear liquid, free from insoluble matter, and shall be free from any defect which may impair its utility.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Inspection.** Sampling for inspection shall be in accordance with provisions of MIL-STD-105, except where otherwise indicated.

4.2.1 *Component and material inspection.* In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3 **Inspection of the end item.**

4.3.1 *Examination of the end item.* The end item and its preparation for delivery shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.2. The lot size shall be expressed in unit containers of the specified capacity for the examinations in 4.3.1.1,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

4.3.1.2, and 4.3.1.3 and in units of shipping containers for the examinations in 4.3.1.4.

4.3.1.1 *Examination of unit container.* The sample unit for this examination shall be one filled unit container.

Examine	Defect
Construction	Not in container specified.
	Any leakage of contents.
	Any split, break, dent, puncture, or other defect.
Closure	Not type specified.
	Any leakage.
Label data	Incomplete; incorrect; missing.
Markings	Illegible; smeared.
	Not in accordance with requirements (see 5-2).

4.3.1.2 *Examination of contents of the unit container.* The sample unit for this examination shall be the contents of one filled unit container.

Examine	Defect
Appearance	Not clear liquid.
	Contains sediment, dirt, or insoluble matter.

4.3.1.3 *Examination of net contents.* The sample shall be conditioned for 10 hours at a temperature of 68°F. prior to examination. The lot shall be unacceptable if the sample data indicates that the average net contents per unit container is less than specified for all sample units examined.

4.3.1.4 *Examination of preparation for delivery.* An examination shall be made to determine that markings, materials, workmanship, and contents comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. This examination shall not apply when the end item container is the shipping container.

Examine	Defect
Markings	Incorrect; illegible; omitted; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Any seepage.
	Any break or dents.

4.3.2 *Inspection levels and AQLs for examination.* The inspection levels for determining the sample size and the AQLs expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.1.1	I	1.0
4.3.1.2	S-4	1.0
4.3.1.3	S-2	—
4.3.1.4	S-2	2.5

4.3.3 *Testing of the end item.* The end item shall be tested for the characteristics specified in table II. The lot size shall be expressed in gallons of insecticide. The sample for testing shall be a one-gallon composite made by selecting an equal amount from the number of containers shown in the sampling table below. The composite sample shall be thoroughly mixed, transferred to a clean, dry, glass container and sealed, labeled, dated, and used for the tests. Duplicate determinations shall be made for each characteristic specified. The lot shall be unacceptable if the composite averages fail to meet any of the test requirements specified. All test reports shall contain the individual values utilized in expressing the final result.

Lot size (gallons)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

TABLE II. *Instructions for testing of the end item*

Characteristic	Specification reference		Requirements applicable to comp. sample	Number determinations per composite sample	Results reported as pass or fail
	Requirement	Test method			
				Average of	
Lindane:					
Melting point	3.1.2	4.4.4	X	2	X
Composition:					
Tetrachloroethylene	3.2	[1]			
Motor oil	3.2	[1]			
Composition by weight DDT	3.2	4.4.2	X	2	X
Composition by weight lindane	3.2	4.4.1	X	2	X
Appearance	3.2.1.1	102[2]	X	2	X
Water content	3.2.1.2	4.4.3	X	2	X

[1] Certificate of compliance will be furnished the contracting officer.

[2] Test performed in accordance to Fed. Test Method Std. No. 791.

4.4 Test procedures.

4.4.1 *Lindane content (by partition chromatography).*

4.4.1.1 *Apparatus and reagents.* The apparatus and reagents to be used are those specified in sections 4.164 and 4.165 of the Official Methods of Analysis of the Association of Official Agricultural Chemists, Ninth Edition (1960).

4.4.1.2 *Preparation of sample.* Weigh at 68°F. 20.0 grams (g.) of the concentrate into a 100-milliliter (ml.) volumetric flask, add 1 ml. of the dye solution and dilute to the mark with the mobile solvent.

4.4.1.3 *Preparation of column.* Transfer 55 g. of silicic acid to the blender, add 165 ml. of mobile solvent and, with mixing, add 30 ml. of nitro-methane. Continue mixing for 15 seconds and pour mixture quickly into the partition column. Wash down column walls with a few ml. of mobile solvent and apply 3 to 4 pounds air pressure to force the liquid into the acid. Allow about 1/2 inch of the liquid to remain over the top of the acid. Insure that the acid column is level; tap, if necessary, to obtain level surface. Release pressure cautiously taking care not to disturb the acid.

4.4.1.4 *Determination.* With the prepared sample at 68°F., pipette 10.0 ml. and allow to flow very slowly down the walls of the tube, taking care not to disturb the surface of the acid. Wash down the tube with mobile solvent and force the solution into the acid by applying 3 to 4 pounds pressure. Release pressure and slowly add (down wall of the column) 150 ml. of the mobile solvent. Reapply pressure, and add 100 ml. more of solvent. Collect the eluent until the solvent obtained is colored with the dye. Set this aside for DDT analysis. Then collect twelve to fifteen 2.5 ml. fractions, transferring each to a separate 50-ml. Erlenmeyer flask and evaporating, as each is obtained, almost to dryness on a steam bath. Allow to cool to room temperature. Examine each flask in order until one or two show crystals. This point indicates the gamma isomer is about to be eluted. Continue to examine the flasks in order for crystal formation. The appearance of a white film that tends to form on the bottom of the flask indicates the gamma component is being obtained. Begin to collect 10 ml. fractions, again transferring each to separate flasks. Continue collection until no crystals are obtained in two successive flasks. This indicates that the total gamma content has been eluted. Add 10 ml. of n-hexane to each flask showing gamma crystals and warm until dissolved. Transfer all solutions quantitatively to a tared 250-ml. flask. Wash by adding 15 ml. of n-hexane

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

to the first 50-ml. flask, transfer washings to each succeeding flask and finally add washings to the tared flask. Repeat washing with another 15 ml. portion of n-hexane. Evaporate on the steam bath until crystallization occurs on cooling. Evacuate the flask for 5 minutes on the vacuum pump, cautiously allowing air to enter the flask several times during this period. Place the flask in a desiccator for 5 minutes. Weigh and calculate the percent lindane in original sample.

4.4.2 *DDT content.*

4.4.2.1 *DDT determination.* Transfer the eluent set aside in the lindane determination to a 250-500 ml. standard tapered Erlenmeyer flask and evaporate to dryness on a steam bath. Add 25 ml. of 99 percent isopropanol and 2.5 g. of metallic sodium in the form of ribbon or cut in small pieces, and shake the flask to mix the sample with the alcohol. Connect to a reflux condenser and boil gently for at least 1/2 hour. Shake the flask occasionally. Eliminate all excess metallic sodium by cautiously adding about 10 ml. of 50 percent isopropanol through the condenser at the rate of 1-2 drops per second. Boil for an additional 10 minutes, add 60 ml. of water through the top of the condenser and then add 5 ml. of 30 percent hydrogen peroxide. Heat the solution to boiling and boil 15 minutes. Disconnect the flask from the condenser and boil for 15 to 30 minutes to remove alcohol. Cool, add 2-3 drops of phenophthalein solution, neutralize by adding nitric acid (1:1) and then add 10 ml. in excess. Add an excess of 15 to 20 ml. of N/10 silver nitrate and coagulate the precipitated silver chloride by digesting on a steam bath for 1/2 hour with frequent stirring. Cool, filter through a Whatman 42 or 44 filter paper, and wash thoroughly with distilled water. Add 5 ml. of saturated ferric-alum solution, and determine the excess silver nitrate in the filtrate by titration with N/10 potassium thiocynate. Subtract the quantity of silver nitrate found in the filtrate from that originally added. The difference will be that required to combine with the chlorine in the DDT. Calculate results as follows:

1 ml. of 0.1*N* $AgNO_3$ = 0.003546 grams Cl

Grams of Cl × 2 = Grams of DDT

Calculate the percent DDT in original sample.

Note: A blank determination should be made following the exact procedure given above but limiting the 0.1*N* $AgNO_3$ solution to 5 ml. in order to obtain a chloride correction for all reagents used.

4.4.3 *Water content.* The water content shall be determined in accordance with method 4082 of Fed. Test Method Std. No. 141.

4.4.4 *Melting point.* The melting point shall be determined by the capillary tube method using a Thiele-Dennis, or equivalent type, melting tube heated at a rate not to exceed 0.5°C. per minute. The temperature at which the sample becomes completely clear shall be the melting point.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 *Levels A and B.* Five gallons of insecticide shall be filled at 68°F. into a five-gallon-capacity pail conforming to type I class 3 of PPP-P-704 with interior coating. Closure shall be screw or snap-on-type and shall be provided with a push-pull pouring spout or a flexible spout. Each closure shall be equipped with a tamper-proof export overseal. The Federal Stock Number shall be embossed on a metal tag and secured to the pail or embossed on the overseal (see 6.3).

5.1.2 *Level C.* The insecticide shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with the rules or regulations of carriers applicable to the mode of transportation.

5.2 Marking. The labeling shall be registered under the Federal Insecticide, Fungi-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

cide, and Rodenticide Act, as amended in compliance with its requirements. The label data shall be durably and legibly marked in accordance with MIL-STD-129, on the main panel of the body by lithographing, stenciling, or printing with a yellow, noncorrosive, weather-resistant, waterproof ink conforming to the requirements of TT-I-558. If the label data are too voluminous to be applied in this manner, directions for use may be placed on a label made of white paper as specified in MIL-STD-129. Labels shall be securely affixed in place, with water-resistant label adhesive as specified in MIL-A-3941 and shall be waterproofed by coating the entire surface of the label with the same adhesive and shall bear the labeling data specified in 3.3.

5.2.1 *Civil agencies.* In addition to labeling specified in 5.2 and any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military requirements.* In addition to labeling specified in 5.2 and any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Insecticide, DDT-Lindane covered by this specification is to be used as specified in equipment which produces space sprays, as a preventive maintenance measure for inside treatment of warehouses and other buildings and to aid in controlling insect infestation.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packing (see 5.1).

6.3 Packing as specified in 5.1.1 is similar to method IA-5 or MIL-P-116.

6.4 Transportation descriptions. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Insecticides, not otherwise indexed by name, other than agricultural.
Carload minimum weight 36,000 pounds.

Motor:

Insecticides, not otherwise indexed other than agricultural.
Truckload minimum weight 36,000 pounds, subject to Rule 34, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL
Navy—SH
Air Force—68

Review activities:

Army—GL, CE, MD, MO, MU
Navy—SH
Air Force—68

User activities:

Navy—MC, MS, WP, YD

Preparing activity:

Army—GL

Review and user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

7

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price ¯ cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

"to detach this form, cut along this line"

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Insecticide, DDT-Lindane Indoor Fogging Type - O-I-561b

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01762

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRE-SS
Natick, Massachusetts 01762

FOLD

"to detach this form, cut along this line"

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.